

Homework 5

1. RECAPITULATION

We start by reminding ourselves of the variables,

$$x_i = \begin{bmatrix} t_1 \\ \vdots \\ t_p \end{bmatrix}, \quad X_{(n,p)} = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix}, \quad \beta_{(p,1)} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad y_{(n,1)} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

In the previous assignment, we differentiated the error function and minimised the derivative to get β ,

$$\beta = (X^T X)^{-1} X^T y \quad \text{when} \quad \frac{\partial (X\beta - y)^T (X\beta - y)}{\partial \beta} = 0.$$

In this assignment, we'll explore β from a more Bayesian point of view.

2. MAXIMUM A POSTERIORI PROBABILITY TECHNIQUE REVIEW

In Bayesian statistics, a *maximum a posteriori probability* (MAP) estimate is an estimate of an unknown quantity that equals the mode of the posterior distribution. Let's say that this distribution has a distribution parameter ζ and (like the normal distribution) the mode, mean and the maximum for this distribution are equal. Fix a new sample point x from the sample's distribution f .

$$x \sim f(\zeta)$$

We may now observe the posterior probability $\mathbf{P}(\zeta | x)$. The idea is that we should reconsider our beliefs of the distribution parameter in the light of new evidence. The evidence being the new sample point x . Suppose that we had only the two choices of $\zeta \in \{\zeta_i, \zeta_j\}$ and post new sample point x we also know that,

$$\mathbf{P}(\zeta_j | x) > \mathbf{P}(\zeta_i | x)$$

Then we should let the new $\zeta = \zeta_j$. More generally,

$$\zeta = \max_i \mathbf{P}(\zeta_i | x)$$

To calculate $\mathbf{P}(\zeta_j | x)$, we use the Bayes' Theorem,

$$\mathbf{P}(\zeta_i | x) = \frac{\mathbf{P}(x | \zeta_i) \mathbf{P}(\zeta_i)}{\mathbf{P}(x)} \quad \text{therefore} \quad \zeta = \max_i \mathbf{P}(\zeta_i | x) = \max_i \frac{\mathbf{P}(x | \zeta_i) \mathbf{P}(\zeta_i)}{\mathbf{P}(x)} = \max_i \mathbf{P}(x | \zeta_i) \mathbf{P}(\zeta_i)$$

The above also allows/requires us to incorporate a prior distribution of ζ since we need it to calculate $\mathbf{P}(\zeta_i)$. Note that x and ζ are not required to be from the same type of distribution. If they are not then $\mathbf{P}(x | \zeta_i)$ and $\mathbf{P}(\zeta_i)$ maybe calculated differently, e. g., have different probability density functions.

3. MAP ESTIMATE TO SOLVE RIDGE REGRESSION

We can drive the normal equation with a regularisation term also known as the solution to the Ridge regression using an MAP estimate of the observed data. We proceed with a linear model having Gaussian noise/error since the data came from the real world.

$$y = x^T \beta + \varepsilon$$

Most of the time, noise ε is modelled as a Gaussian random variable.

$$\varepsilon \sim \mathcal{N}(0, c^2), \quad \text{for } c \in \mathbb{R}$$

This is justified since noise maybe caused by numerous little factors, all adding up to a Gaussian due to the central limit theorem. We further assume $\beta \in \mathbb{R}^p$ to have a prior Gaussian distribution centred around 0 with a variance of $\tau^2 \mathbb{I}_p$: $\beta \sim \mathcal{N}(0, \tau^2 \mathbb{I}_p)$. Due to the above modelling of noise and β , we know that y are also Gaussian and with expected values as a linear function of x . We state our assumptions on the data:

Normal Distribution: All y_i are normally distributed: $y_i \sim \mathcal{N}(\mu_i, \sigma^2)$.

Variance: We assume that all y_i have the same variance σ^2 .

Linearity: We assumed that there is indeed a linear relationship between $x_i^T \in X_{(n,p)}$ and $y_i \in \mathbb{R}$. This is mathematically expressed as: the expected or mean value of y_i is a linear function of x_i ,

$$\mathbb{E}[y_i|x_i] = \mu_i = x_i^T \beta.$$

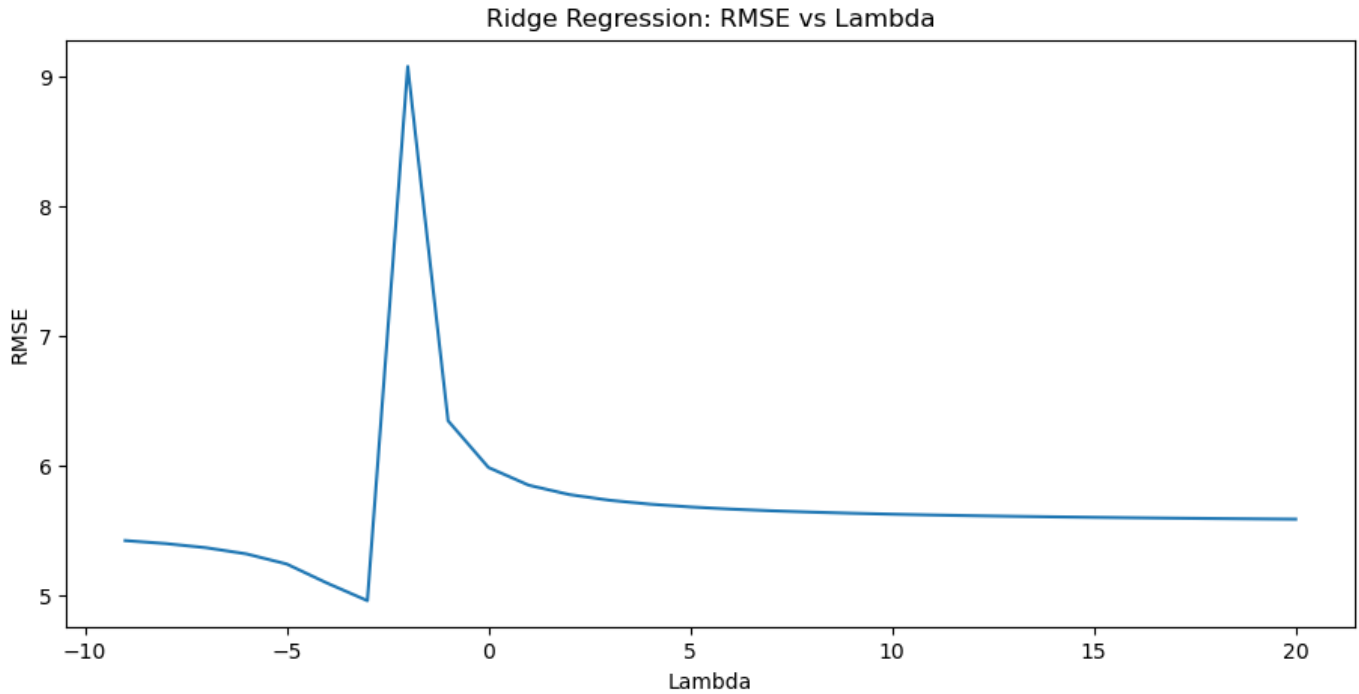
Independent & Identically Distributed (IID): We assume that all y_i are independent and are from the same type of distribution.

State the problem of finding the best β as finding the most probable *posterior* $\mathbf{P}(\beta | y_i)$,

$$\beta = \max_{\beta} \mathbf{P}(\beta | y_i) \quad \text{and due to the Bayes' Theorem,} \quad \mathbf{P}(\beta | y_i) = \frac{\mathbf{P}(y_i | \beta) \mathbf{P}(\beta)}{\mathbf{P}(y_i)}$$

By the IID assumption,

$$\begin{aligned} \mathbf{P}(y | \beta) &= \prod_{i=1}^n \mathbf{P}(y_i | \beta) \\ &= \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 \right) \quad \text{PDF of } \mathcal{N}(\mu_i, \sigma^2) \\ \max_{\beta} \mathbf{P}(\beta | y) &= \max_{\beta} \frac{\mathbf{P}(y | \beta) \mathbf{P}(\beta)}{\mathbf{P}(y)} \\ &= \max_{\beta} \frac{\prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 \right) \mathbf{P}(\beta)}{\mathbf{P}(y)} \\ &= \max_{\beta} \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 \right) \mathbf{P}(\beta) \\ &= \max_{\beta} \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 \right) \frac{1}{\tau \sqrt{2\pi}} \exp \left(-\frac{1}{2\tau^2} (0 - \beta)^2 \right) \\ &= \max_{\beta} \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 \right) \frac{1}{\tau \sqrt{2\pi}} \exp \left(-\frac{1}{2\tau^2} \beta^2 \right) \\ &= \max_{\beta} \prod_{i=1}^n \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 \right) \exp \left(-\frac{1}{2\tau^2} \beta^2 \right) \\ &= \max_{\beta} \prod_{i=1}^n \exp \left(-\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 - \frac{1}{2\tau^2} \beta^2 \right) \\ &= \min_{\beta} \prod_{i=1}^n \exp \left(\frac{1}{2\sigma^2} (x_i^T \beta - y_i)^2 + \frac{1}{2\tau^2} \beta^2 \right) \\ &= \min_{\beta} \prod_{i=1}^n \exp \left((x_i^T \beta - y_i)^2 + \frac{2\sigma^2}{2\tau^2} \beta^2 \right) \\ &= \min_{\beta} \prod_{i=1}^n \exp \left((x_i^T \beta - y_i)^2 + \lambda \beta^2 \right) \quad \lambda = \frac{\sigma^2}{\tau^2} \\ &= \min_{\beta} \sum_{i=1}^n \ln \exp \left((x_i^T \beta - y_i)^2 + \lambda \beta^2 \right) \\ &= \min_{\beta} \sum_{i=1}^n (x_i^T \beta - y_i)^2 + \lambda \beta^2 \\ &= \min_{\beta} (X\beta - y)^T (X\beta - y) + \lambda \beta^T \beta \end{aligned}$$



At this point, in order to minimise the above, we differentiate $(X\beta - y)^T(X\beta - y) + \lambda\beta^T\beta$ and set it equal to zero. From the previous assignment, we know that,

$$\frac{\partial(X\beta - y)^T(X\beta - y)}{\partial\beta} = 2X^T X\beta - 2X^T y \quad \text{and} \quad \frac{\partial\lambda\beta^T\beta}{\partial\beta} = 2\lambda\beta$$

Therefore, the derivative equal to zero is $2X^T X\beta - 2X^T y + 2\lambda\beta = 0$,

Question 1. Let $\mathbb{I}$ be a p by p identity matrix. Solve $2X^T X\beta - 2X^T y + 2\lambda\beta = 0$ for β as the new linear model. Show your work.

$$\begin{aligned} X^T X\beta - X^T y + \lambda\beta &= 0 \\ X^T X\beta + \lambda\beta &= X^T y \\ (X^T X + \lambda\mathbb{I})\beta &= X^T y \\ \beta &= (X^T X + \lambda\mathbb{I})^{-1} X^T y \end{aligned}$$

Question 2. Using the same two columns you picked in the last assignment from Boston house-price data, draw a (appropriately labelled) plot using the equation you got in question 1. The plot should graph integers $-10 < \lambda < 20$ on the x -axis and the model's root mean squared error on the y -axis. Use a file called `ridge.py` for this question.

Question 3. From your graph, what is the best value of the hyper-parameter λ that corresponds to the smallest root mean squared error? Give both the best $-10 < \lambda < 20$ and the corresponding error.

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SUBMISSION INSTRUCTIONS

- 1) Submit a PDF that answers the questions and contains all the plots that the assignment asks for.
- 2) Submit your `ridge.py`.